

rl-matdesign — YAML Configuration Reference

Every config flag, what it does, and its choices

2026-06-10

Overview

A run is configured by one YAML file plus a few CLI flags. The three pluggable layers are chosen by **env_type** (the search space), **predictor** (the reward), and **constraint_filter** (action masking); each then reads its own keys. The RL and generation knobs are shared across scenarios.

- The YAML is the base; a few CLI flags override matching keys.
- **Seeds are CLI-only:** `--dp-seed` (predictor RNG), `--train-seed` (training; defaults to `--dp-seed`), `--gen-seed` (generation; defaults to `--dp-seed`).
- The full merged config is written to `<out>/run_config.json`.

CLI flags: `--config` (required), `--out` (required), `--method`, `--dp-seed`, `--train-seed`, `--gen-seed`, `--device` (cpu/cuda), `--dqn-loss`, `--dqn-augment-permutations`, `--save-checkpoint-freq`, `--resume-training`, `--only-generate`, `--skip-generation`, `--max-gen-attempts`, `--load-qnet` / `--load-policy` / `--load-value-net` / `--load-scaler`.

1. Environment

Key	What it does	Choices / default
env_type	Selects the environment	<code>fraction</code> (default, <code>CompositionEnv</code>); <code>integer_ratio</code> (<code>IntegerRatioEnv</code>); <code>multi_group</code> (<code>MultiGroupEnv</code>)

fraction env keys

Key	What it does	Choices / default
cation_set	Candidate element symbols	required (list)
fraction_set	Allowed fraction strings (grid)	default = built-in <code>0.05 ... 0.80</code>
total_units	Grid resolution; step = $1/\text{total_units}$	<code>20</code> ($\rightarrow 0.05$); use <code>100</code> for <code>0.01</code>
n_components	Distinct elements per episode	<code>5</code>
anion_formula	Fixed suffix appended to the formula	<code>""</code> (e.g. <code>02H1</code> , <code>03</code>)
episode_style	How each step picks	<code>element_then_amount</code> (default: choose element + amount); <code>fixed_order_amount</code> (order pinned to <code>cation_set</code> , choose only amounts)
element_bounds	Per-element <code>{el: [min,max]}</code> fraction caps	<code>none</code> ; only with <code>fixed_order_amount</code>
constraint_filter	Action-masking filter (see §3)	<code>null</code>

integer_ratio env keys

Key	What it does	Choices / default
cation_set	Candidate elements	required
ratio_set	Allowed integer-ratio digits	default <code>"0" ... "9"</code>
n_components	Elements per episode	<code>5</code>
constraint_filter	Action-masking filter	<code>null</code>

No sum-to-1 constraint; the composition is normalized afterward.

2. multi_group env — groups:

groups: is an **ordered list** of sublattice **groups**, filled in order. A group is the atomic unit; a single fraction/integer_ratio env is the one-group case. The predictor receives the structured {group_name: {element: value}}.

Common group keys

Key	What it does	Choices / default
name	Group label (appears in the structured terminal)	string
kind	Group type	composition (default) or categorical
sites	Sublattice size (atoms per formula unit) — bridges fraction<->count and assembles the chemical formula (amount × sites)	1
constraint_filter + its kwargs	Per-group filter (sees prior_groups for cross-group coupling)	(see §3)

kind: composition — pick N elements with amounts summing to 1. Reuses the fraction-env keys (cation_set, fraction_set, total_units, n_components, episode_style, element_bounds) plus friendly knobs:

Key	What it does	Choices / default
amount	Generate the value grid instead of writing fraction_set	{min, max, step} or a list
host	A host element auto-takes the leftover (list only the dopants; wires a host_complement filter)	element symbol

kind: categorical — pick discrete **real values**; no sum-to-1. Each slot is one element with its own value list; the terminal returns the chosen values unchanged (so a builder reads $Cl = 1.0 / 0 = 1$ directly).

Key	What it does
choices	[{element, values: [...]}], ...] — one slot per element

A categorical filter may mask a slot by an earlier group (e.g. sse_doping masks the O slot by the P-site metal's category).

3. Constraint filters (constraint_filter:)

Value	What it does	Its keys
null	No constraint (default)	—
smact_charge	Keep only compositions that can be charge-neutral (SMACT oxidation states)	smact_anions: [{symbol, charge, stoich}] (or smact_anion / smact_anion_charge / smact_anion_stoich)

Value	What it does	Its keys
last_step_element	Force the last step to a required element	required_elements; nonzero_ratio_at_last (true); reserve_for_last (true)
phase_pattern	Keep compositions matching phase patterns	phase_patterns
ooh_phase chain	OOH oxyhydroxide phase screen Apply several filters in sequence	target_phases (default [any]) filters: [{constraint_filter: ..., ...}, ...]
host_complement	Dopants at a level, host takes the rest (wired automatically by a composition group's host knob)	host_element, levels
sse_doping	Mask the LiPS S-site O-form slot by the metal's category	o_element (O), host_P (P), metal_only, oxide_only
"pkg.mod:Class"	Your own filter (FQN)	your class's keys

4. Predictors (predictor:)

Value	What it does
dummy	Random reward (default; no models — smoke tests)
dp_structure	Substitute placeholder sites -> DP energy ensemble
dp_property	Substitute -> DP property-vector ensemble (DeepProperty)
composite	Weighted combo of child predictors (shares the <i>composition</i>)
structure_pipeline	Build once -> relax once -> N property ensembles (shares the <i>structure</i>)
ooh	OOH overpotential (adsorbate slabs)
hea / perovskite	dp_structure subclasses (different default site_symbol)
sinter_calcine	RandomForest sintering/calcine temperature
"pkg.mod:Class"	Your own predictor (FQN)

dp_structure / dp_property keys

Key	What it does	Default
base_poscar (or poscar / poscar_template)	Template with placeholder sites	required
dp_models	List of .pt checkpoints	required
site_symbol	Placeholder element	X (hea/perovskite differ)
dp_head	Model head	none / property (dp_property)
objective	(mean,std) -> scalar	mean_minus_kstd; also mean, mean_plus_kstd
k	Uncertainty coefficient	1.0
n_random_configs	Random placements averaged	5
energy_per_atom (dp_structure)	Normalize by atom count	true
output_index (dp_property)	Which vector component	0
output_aggregator (dp_property)	Collapse vector -> scalar	index (index/mean/max)
maximize (dp_property)	Larger raw value = better	false

composite keys

objectives: [{name, predictor, direction (min/max), weight, scale, objective}, ...], plus k. Shared base_poscar / site_symbol / n_random_configs inherit into children unless a child overrides them.

structure_pipeline keys

Key	What it does	Default
builder	Builder name/FQN (composition/groups -> ASE Atoms)	required (e.g. sse)
n_random_configs	Placements per candidate	1
geo_opt	Relaxation stage (see below); omit / enabled: false to skip	optional
properties	List of property specs (see below)	required
k	Uncertainty coefficient	1.0

geo_opt: sub-keys — model (default models/DPA-3.1-3M.pt), head (user-defined), fmax (0.001), steps (1000), relax_cell (true), enabled (true).

properties[]: sub-keys — name, models (ensemble list), head, direction (max/min), weight (1.0), scale (1.0), objective (mean_minus_kstd/mean/mean_plus_kstd), output_index (0), output_aggregator (index).

Reward = $\sum \text{weight} * (\text{direction} * \text{mean} - k * \text{std}) / \text{scale}$ over the properties.

ooh keys

base_poscar, dp_models, objective, k, n_random_configs, ads_height, ads_dz, geo_opt (bool), geo_opt_model, uncertainty (models/configs/total), output_index, target_phases.

sinter_calcine keys

rf_model (joblib path), mode (sinter / calcine).

5. Builder (builder: sse) keys

Key	What it does	Default
base_poscar	Base supercell	required
valences	Charge table {el: int \ {sulfide, oxide}} (drives the Li-vacancy solve and oxide O count)	required
halide_total	Cl + Br per formula unit	1.7
eligible_region	Substitutable S region	{symbol: S, take: last, count: 1000}
formula_units	Formula units in the supercell — inferred from the POSCAR (host-P count / p_site_per_fu) unless set	inferred
p_site_group / s_site_group	Group names to read	P_site / S_site
host	{P, S, Li} host symbols	{P:P, S:S, Li:Li}
p_site_per_fu / s_site_per_fu / li_per_fu	Sites per formula unit	1 / 6 / 6

The S-site reads the categorical values directly — O is a form flag (0 metal / >0 oxide), Cl is a real per-f.u. count. Br = halide_total - Cl and the charge-neutral Li vacancy are derived. (No cl_map / o_off.)

6. RL method & hyperparameters

Key	What it does	Choices / default
method	Algorithm (CLI --method overrides)	a2c (default), reinforce, dqn

Policy gradient (a2c / reinforce)

Key	Default	Key	Default
pg_warmup_eps	200	pg_entropy_coef	0.01
pg_num_iters	1000	pg_repeat_penalty_coef	0.0
pg_batch_eps	15	pg_repeat_penalty_shape	log (also sqrt)
pg_lr_actor	0.001	gamma (or pg_gamma)	0.9
pg_lr_critic	0.001 (a2c only)		

DQN

Key	Default	Key	Default
dqn_lr	0.001	dqn_target_update_freq	100
dqn_hidden_dim	256	dqn_eps_anneal_eps	10000
dqn_batch_size	256	dqn_eps_min	0.05
dqn_warmup_eps	->pg_warmup_eps/500	dqn_gamma (or gamma)	0.9
dqn_num_train_eps	20000	dqn_loss	smoothl1 (also mse)
dqn_buffer_size	50000	dqn_augment_permutation	0
dqn_grad_steps_per_ep	5		

Generation (all methods)

Key	What it does	Default
num_gen_eps	Final candidates to generate	200
exploration_gen_eps	Extra high-temperature candidates	0
gen_temperature	Sampling temperature (DQN Boltzmann / PG)	1.0
gen_top_frac	Restrict sampling to top fraction of actions	0.0
gen_epsilon	ϵ -random during generation	0.0
k	Uncertainty coefficient for reward folding	1.0